

TINA REZVANIAN

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SUMMARY

Applied ML Scientist with 7+ years developing generative AI and large-scale, user-facing products, including Adobe Firefly, Liquid Mode, and Adobe Scan. Led core contributions to the development, fine-tuning, and evaluation of Firefly's diffusion-based image generation models, advancing productionization, image quality, inpainting reliability, and prompt controllability. Strong track record in image generation and vision research, with cross-functional collaboration across research, product, and design teams to translate creative workflows into reliable, controllable generative tools.

EDUCATION

Northeastern University Ph.D., Industrial Engineering and Operations Research Minor in Mathematics	Boston, MA 2013 - 2019
Northeastern Illinois University MBA, Master of Business Administration	Chicago, IL 2011 - 2013
University of Science and Technology of Mazandaran B.Sc., Industrial Systems Engineering	Mazandaran, Iran 2006 - 2010

SKILLS

AI/ML: Diffusion Models, Vision-Language Models (CLIP/BLIP), Transformers, VAEs, Temporal Consistency Methods, Reinforcement Learning, Causal Models, Representation Learning, Ensemble Models, Active Learning, Dimensionality Reduction, Clustering, Similarity Search Auto-Labeling, Graph Models, Normalization Regularization, Advanced Optimization, Mixed-Integer Stochastic Programming

Programming & ML Systems: Python, PyTorch, TensorFlow, scikit-learn, SQL, Docker, Kubernetes, AWS, Hadoop/Spark, RunAI, CI/CD Pipelines, CPLEX, Gurobi

Visualization: Matplotlib, Plotly, Seaborn, Tableau, Power BI, Gephi.

WORK EXPERIENCE

Adobe <i>Machine Learning Engineer, Data Scientist, Firefly</i>	Sep 2019 - May 2024 <i>San Jose, CA</i>
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- Firefly: Developed and finetuned multiple Firefly's modules such as input/output data encoder, input dense vector retrieval, diffusion Unet, and in-painting inference
 - . Enriched data with generated captions, GenAI images, and multi-representational embeddings.
 - . Developed evaluation models to assess coherence and technical quality of images using human-scores
- AB Testing Firefly: Evaluated the impact of data enrichment strategies and architectural modifications to address model limitations and enhance performance.
 - . Developed a rapid experimentation framework automating training and evaluation; integrating RunAI, AWS, Artifactory; streamlining job configurations, checkpoints, Docker, GPU, and data transfer.
- Gap Analysis: Identified subjects, contexts, and actions where Gen AI performance was suboptimal.
 - . Resolved data issues in text presence, aesthetic scoring, face/human detection, and captioning.
 - . Extracted insights from unsupervised similarity, categorization, caption alignment, and segmentation.
 - . Optimized caption weights by NLP analysis, focusing on alignment, part of speech, and model biases.

- Multi-Channel Attention Attribution: Boosted paid user retention and trial conversion by recommending next-best action through estimating the credit share of each touchpoint and its time-of-use.
 - . Trained a scalable multi-head attention model with billions of parameters, visualized in a web app.
 - . Identified critical workflows for engagement during the user journey
- Super Users: Segmented and characterized users to recommend workflow sequences, optimizing retention and personalizing onboarding, in-app, and email communications.
 - . Trained a β -variational auto-encoder to automatically extract meaningful attributes
 - . Generated similar sequences to that of retained users using the decoder within attribute-based clusters
- FontMART: Ranked fonts to maximize reading speed using LambdaMART, directly optimizing NDCG.
 - . Gathered and evaluated data from MTurk, web apps, and Hadoop to ensure quality and readiness.
 - . Identified optimal fonts, fastest speeds, and key font traits by age group.
- RealTime Recommender: Developed an on-device real-time recommender using reinforcement learning, trained via interactions with a generative AI user agent.
 - . Trained a generative adversarial network to model generalized user behavior from sequential data.
 - . Built a general deep Q-learning recommender and personalized it through user-specific fine-tuning.
- CausalTIM: Performed causal analysis to identify features' impact on conversion using observational data.
 - . Developed a parallel workflow for propensity score matching across features, handling data preparation, instance matching, and treatment effect estimation.

Harvard Extension School

Nov 2019 - May 2020

- Face Recognition: Built face recognition models using CNNs on the CASIA-WebFace dataset, adapting FaceNet, DeepFace, VGGFace, and ResNet through transfer learning, fine-tuning, and data augmentation

Insight Data Science

Jan 2019 - Feb 2019

- PickaBasket: Created a Python/Flask app on AWS that recommends bestseller grocery bundles using graph algorithms and probabilistic co-purchase patterns.

Wheels UP Private Jet Company

May 2018 - Aug 2018

Data Science Intern

New York City, NY

- Forecasted daily flight demand and allocated aircraft to regions to minimize empty flights, clustering movements with regional metrics and community detection methods.

Northeastern University

Dec 2014 - Sep 2019

Research Assistant

Boston, MA

- Forecasted highly uncertain, zero-inflated emergency demand data for UNHCR using a two-stage gradient boosting approach that minimized calibration error in the first stage and applied label smoothing in the second. Outperformed tree-based ensembles, Tweedie, and Poisson models on skewed distributions.
- Integrated emergency and ongoing networks in a multi-objective framework, reducing cost by 31% and response time by 18% with a two-stage stochastic model, delivering a Python-based decision support.
- Introduced a novel matching algorithm to solve relaxed-stable matching in multi-period, incomplete graphs by extending the Top Trading Cycle mechanism with negotiated exchanges. Applied to a World Food Programme case and deployed as a Python-based decision support app.

Healthcare Systems Engineering Institute (HSYE)

Sep 2013 - Dec 2014

Northeastern University, Research Assistant

Boston, MA

- Scheduled Treatments and Nurses using Column Generation Algorithm at Moffit Cancer Center.
- Optimized Curbside Threshold at Massachusetts General Hospital (MGH) Neurology Department.

PUBLICATIONS

- . Beier, Sofie, et al. "Readability research: An interdisciplinary approach." Foundations and Trends® in Human-Computer Interaction 16.4 (2022): 214-324.
- . Cai,T. , Wallace,S., **Rezvanian,T.**(2022, June). Personalized Font Recommendations:Combining ML and Typographic Guidelines to Optimize Readability. In Designing Interactive Systems Conference(pp.1-25)
- . Jahre,M.,Kembro,J., **Rezvanian,T.**, Ergun.O, Hapnes(2016)*Integrating supply chains for emergencies and ongoing operations in UNHCR*, Journal of Operations Management,45,57-72.
- . Jahre, M., Kembro, J., **Rezvanian, T.**, Ergun, O., Haapnes, S.,& Hultberg, M. *Scenario building in UNHCR-Predicting future demand in refugee crises*. In Proceedings of the 5th World Production and Operations Management Conference, POM Havana 2016
- . **Rezvanian, T.**, Ergun, O., Jahre, M., Kembro, J., (2020) *Forecasting Emergency Response Demand for Refugees as a Zero-Inflated Skewed variables: Evidence from UNHCR*
- . **Rezvanian, T.**, Ergun, O., *Two-sided Stable Matching with Negotiations*, Integrating Data-Driven Forecasting and Large-Scale Optimization to Improve Humanitarian Response Planning and Preparedness. Chapter 1, Diss. Northeastern University,
- . **Rezvanian, T.**, Ergun, O., *Deferred Acceptance Algorithms and Aftermarkets for Matching with Short Preference Lists at World Food Programme*, Integrating Data-Driven Forecasting and Large-Scale Optimization to Improve Humanitarian Response Planning and Preparedness. Chapter 1, Diss. Northeastern University, 2019.